Tejaswaroop Reddy Yanamala

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EXPERIENCE

Edunet Foundation Jun 2025 – Aug 2025

Full-Stack Developer Intern | *React.js, Node.js, Express.js, MongoDB, Redis, Docker, AWS EC2/S3, Nginx* Live Demo

- Architected a full-stack civic issue reporting platform with **12 REST APIs**, role-based authentication, and a 2-schema MongoDB data model, deployed with Docker on **AWS EC2** behind Nginx
- Implemented **Redis** cache-aside caching, **reducing median data-retrieval latency from 87.08ms to 1.42ms (~98% reduction)** across 35 benchmark runs, with cache invalidation verified on issue mutations
- Engineered geospatial duplicate detection using MongoDB **2dsphere** indexing + $near queries, achieving **41.44ms median query latency** across 35 runs for 100m-radius proximity checks
- Secured images with private AWS S3 storage, 1-hour presigned URLs, and EC2 IAM roles instead of embedded credentials

PROJECTS

CareerPilot – AI Interview-Prep Copilot | *React, FastAPI, LangChain, LangGraph, MCP, MongoDB, Docker*

- Built an **AI interview-prep copilot** with **3 LangGraph state graphs (12 nodes)**, MongoDB checkpointing, and human-in-the-loop interrupts for plan confirmation and clarification
- Designed **hybrid RAG retrieval** over 3 knowledge bases using dense (bge-base) + BM25 fused with RRF and a cross-encoder reranker, achieving **hit@5 of 1.0 and MRR of 0.96** on a 33-query eval set
- Built 2 bounded **ReAct agents** calling a custom **FastMCP** server for live job search, GitHub repo ingestion, and application tracking; job ranking **precision@3 of 0.96** on 40 labelled pairs
- Added grounding guardrails, PII stripping, and prompt-injection defences (**0 of 4 attacks obeyed**); validated with **80 automated tests** and a LangSmith eval harness; deployed with Docker Compose

PayBridge AI – Razorpay Debugging Platform | *Node.js, React, MongoDB, Razorpay, LangGraph, MCP*

- Built a **Razorpay** debugging platform with a **rules-first diagnosis engine** (LLM only for unclear cases) that finds root causes from webhooks and API errors; validated on **21 failure scenarios**
- Developed a **LangGraph triage agent** with **6 read-only MCP tools** scoped to one merchant, so the model cannot see other merchants' data or change payments
- Added **hallucination control** (validator checks every AI claim against retrieved records), PII redaction, and prompt-injection defence
- Secured **webhooks** with **HMAC-SHA256 verification**; built a **REST API** with token auth and rate limiting, covered by **118 backend and 33 frontend tests**

Cache Replacement Policy Simulator | *C++, STL, DSA, OOP*

- Implemented FIFO, LRU, and LFU eviction in C++ with STL containers, achieving **O(1) operations** for LRU and LFU
- Created a workload generator simulating realistic 80/20 hot-cold memory access patterns to benchmark policy performance
- Achieved **80.61% hit rate with LFU vs. 57.55% with FIFO** on a 20,000-access workload, a 40% relative improvement

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL

Web & Backend: React.js, Tailwind CSS, Node.js, Express.js, FastAPI, REST APIs, Webhooks, Razorpay

AI / LLM: LangChain, LangGraph, Model Context Protocol (MCP), FastMCP, RAG, Hybrid Retrieval, ReAct Agents, Agent Guardrails, Hallucination Control, Prompt-Injection Defence

Databases: MongoDB, MySQL, Redis

Cloud & Tools: AWS (EC2, S3, IAM), Docker, Nginx, Git, GitHub, Postman, cURL, Debugging & Root-Cause Analysis, Technical Documentation

Core: DSA, OOP, DBMS, Operating Systems, Computer Networks

EDUCATION

SRM University AP Aug 2023 – May 2027

B.Tech in Computer Science and Engineering – CGPA: 8.90 Guntur, AP

Sri Chaitanya Junior College Aug 2021 – Mar 2023

Class XII – 95.7% Tirupati, AP

ACHIEVEMENTS AND CERTIFICATIONS

- **AWS Certified AI Practitioner** (Exam Code: AIF-C01)
- Solved **300+ DSA problems** across LeetCode, GeeksforGeeks, and TakeUForward